

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2024

TEST DATE: 28-12-2023

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

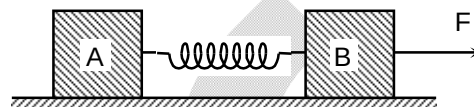
Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. Two bodies A and B each of mass M are fixed together by a massless spring. A force F acts on the mass B as shown in figure. At the instant shown the mass A has acceleration a . What is the acceleration of mass B?



- (A) $(F/M) - a$
(B) a
(C) $-a$
(D) (F/M)

2. Potential energy of a body of mass ' m ' in a conservative force field can be expressed $U = \alpha x - \beta y$ as where x and y are position coordinates of the body. Acceleration of the body can be expressed as:

(A) $\frac{\alpha^2 - \beta^2}{m}$

(B) $\left[\frac{\alpha^2 + \beta^2}{m^2} \right]^{1/2}$

(C) $\frac{\alpha - \beta}{m}$

(D) $\frac{\alpha + \beta}{m}$

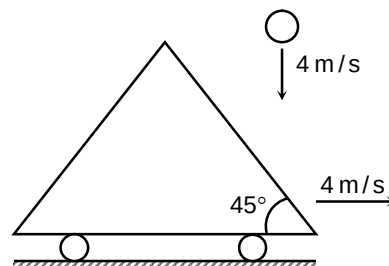
3. A small ball falling vertically downward with constant velocity 4 m/s strikes elastically a massive inclined cart moving with velocity 4 m/s horizontally as shown. The speed of the rebound of the ball is

(A) $4\sqrt{2} \text{ m/s}$

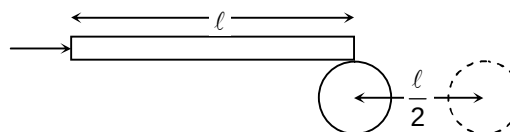
(B) $4\sqrt{3} \text{ m/s}$

(C) 4 m/s

(D) $4\sqrt{5} \text{ m/s}$



4. A cylindrical drum is pushed along by a board of length l . The drum rolls forward on the ground a distance of $\frac{l}{2}$. There is no slipping at any instant.



During the process of pushing the board, the distance moved by the man on the ground is:

(A) $\frac{l}{2}$

(B) $\frac{3l}{4}$

(C) l

(D) none of these

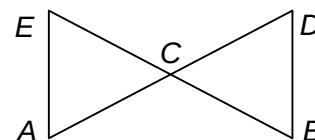
5. If resistance of each wire in the network shown is r , the equivalent resistance between A & C is equal to

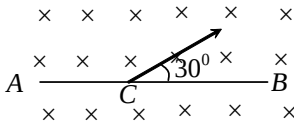
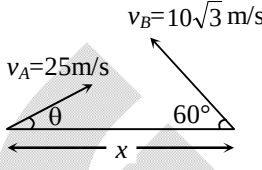
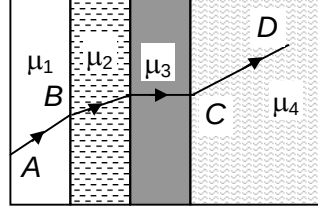
(A) r

(B) $\frac{r}{2}$

(C) $\frac{2r}{3}$

(D) $\frac{3r}{2}$



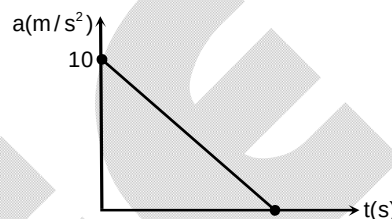
6. A conducting rod AB of length $l = 1\text{ m}$ is moving at a velocity $v = 4\text{ m/s}$ making an angle 30° with its length. C is the middle point of the rod. A uniform magnetic field $B = 2\text{ T}$ exists in a direction perpendicular to the plane of motion, then
- (A) $V_A - V_B = 8\text{ V}$ (B) $V_A - V_B = 4\text{ V}$
 (C) $V_B - V_C = 2\text{ V}$ (D) $V_B - V_C = 4\text{ V}$
- 
7. Two particles A and B are projected simultaneously in the directions shown in figure with velocities $v_A = 25\text{ m/s}$ and $v_B = 10\sqrt{3}\text{ m/s}$. If they collide in air after 2 s , the angle θ is
- (A) 30° (B) 45°
 (C) 53° (D) 37°
- 
8. Three capillaries of length L , $L/2$ and $L/3$ are connected in series. Their radii are r , $r/2$ and $r/3$ respectively. Then if stream-line flow is to be maintained and the pressure across the first capillary is P , then
- (A) the pressure difference across the ends of second capillary is $8P$
 (B) the pressure difference across the third capillary is $81P$
 (C) the pressure difference across the ends of the second capillary is $16P$
 (D) the pressure difference across the third capillary is $9P$
9. An electron in hypothetical hydrogen atom is in its 3^{rd} excited state and makes transition from 3^{rd} to 2^{nd} excited, then to 1^{st} excited state and then to ground state. If the amount of time spent by the electron in any state of quantum number n , is proportional to $\left(\frac{1}{n-1}\right)$, then the ratio of no. of revolutions completed by the electron in 1^{st} excited state to that in the 2^{nd} excited state will be
- (A) 2 (B) $\frac{27}{8}$
 (C) $\frac{27}{4}$ (D) $\frac{27}{6}$
10. A ray of light passes through four transparent media with refractive indices μ_1 , μ_2 , μ_3 and μ_4 as shown in the figure. The surface of all media are parallel. If the emergent ray CD is parallel to the incident ray AB , then
- (A) $\mu_1 = \mu_2$ (B) $\mu_2 = \mu_3$
 (C) $\mu_3 = \mu_4$ (D) $\mu_4 = \mu_1$
- 
11. The mass of the moon is $\frac{1}{81}$ of the earth but the gravitational pull is $\frac{1}{6}$ of the earth. It is due to the fact that
- (A) The radius of the moon is $\frac{81}{6}$ of the earth (B) The radius of the earth is $\frac{9}{\sqrt{6}}$ of the moon
 (C) Moon is the satellite of the earth (D) None of these
12. The potential energy of a particle of mass 1 kg in motion along the x -axis is given by $U = 4(1 - \cos 2x)\text{ J}$. Here x is in metres. The period of small oscillations (in sec) is
- (A) 2π (B) π
 (C) $\frac{\pi}{2}$ (D) $\sqrt{2}x$

13. For $\frac{1}{m}$ of the distance between two stations a train is uniformly accelerated and $\frac{1}{n}$ of the distance it is uniformly retarded. It starts from rest at one station and comes to rest at the other. The ratio of the greatest velocity of the average velocity will be:

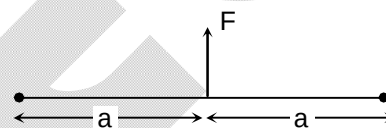
(A) $\left(1 + \frac{1}{m} - \frac{1}{n}\right)$ (B) $\left(1 + \frac{1}{m} + \frac{1}{n}\right)$
 (C) $\left(\frac{1}{m} + \frac{1}{n} - 1\right)$ (D) $\left(\frac{1}{m} + \frac{1}{n}\right)$

14. A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be

(A) 110 m/s (B) 55 m/s
 (C) 550 m/s (D) 660 m/s



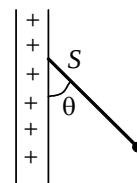
15. Two particles of mass 'm' each are tied at the ends of a light string of length '2a'. The whole system is kept on frictionless horizontal surface with the string held tight so that each mass is at a distance 'a' from the centre P (as shown in figure). Now, the mid-point of the string is pulled vertically upwards with a small but constant force F. As a result, the particles move towards each other on the surface. The magnitude of acceleration, when the separation between them becomes 2x is:



(A) $\frac{F}{2m} \frac{a}{\sqrt{a^2 - x^2}}$ (B) $\frac{F}{2m} \frac{x}{\sqrt{a^2 - x^2}}$
 (C) $\frac{F}{2m} \frac{x}{a}$ (D) $\frac{F}{2m} \sqrt{\frac{a^2 - x^2}{x}}$

16. A charged ball hangs from a silk thread of length l. It makes an angle θ with a large charged conducting sheet P as shown in the figure. The surface charge density σ of the sheet is proportional to

(A) $\cos \theta$ (B) $\cot \theta$
 (C) $\sin \theta$ (D) $\tan \theta$

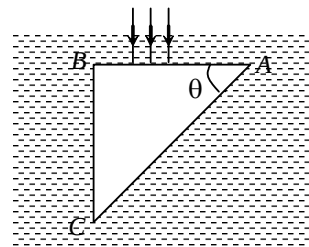


17. What is the ratio of the circumference of the first Bohr orbit for the electron in the hydrogen atom to the de-Broglie wavelength of electrons having the same velocity as the electron in the first Bohr orbit of the hydrogen atom?

(A) 1 : 1 (B) 1 : 2
 (C) 1 : 4 (D) 2 : 1

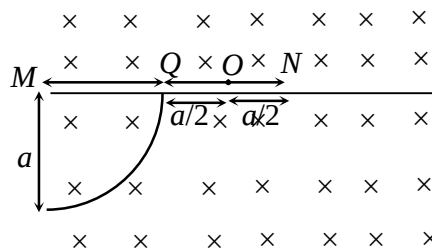
18. A glass prism of refractive index 1.5 is immersed in water ($\mu = 4/3$). A light beam incident normally on the face AB is totally reflected to reach the face BC, if

(A) $\sin \theta \geq \frac{8}{9}$ (B) $\sin \theta \leq \frac{2}{3}$
 (C) $\sin \theta = \frac{4}{5}$ (D) $\frac{2}{3} < \sin \theta < \frac{8}{9}$



19. There is a uniform magnetic field perpendicular to the plane of paper. A positive charged particle enters in the region, perpendicularly and collides inelastically at point Q to the rigid wall MN . If Co-efficient of restitution $e = 1/2$ then particle

(A) Retraces its path
(B) Strikes at point O
(C) Strikes at point M
(D) strikes at point N



20. A tuning fork of frequency 340 Hz is vibrated just above a cylindrical tube of length 120 cm. Water is slowly poured in the tube. If the speed of sound is 340 m/s then the minimum height of water required for resonance is

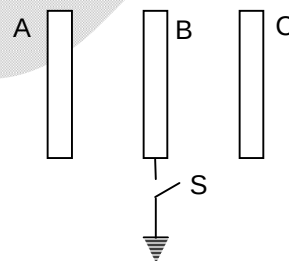
(A) 25 cm
(B) 45 cm
(C) 75 cm
(D) 95 cm

SECTION – B

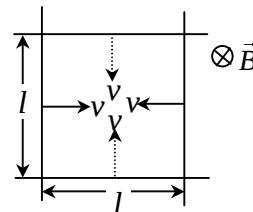
(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

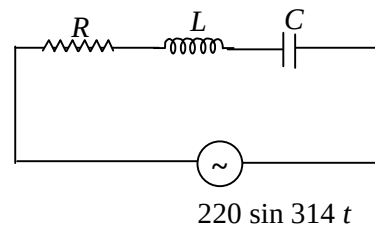
21. Three parallel conducting plates (A, B, C) are given initial charges Q , $2Q$, $4Q$ respectively. If the switch S is closed, the charge flowing to earth is xQ . Value of x is



22. In the figure shown the four rods have $\lambda = 0.5 \Omega/\text{m}$ resistance per unit length. The arrangement is kept in a magnetic field of constant magnitude $B = 2\text{ T}$ and directed perpendicular to the plane of the figure and directed inwards. Initially the rods form a square of side length $l = 15\text{ m}$ as shown. Now each wire starts moving with constant velocity $v = 5\text{ m/s}$ towards opposite wire. find the force required in newton on each wire to keep its velocity constant at $t = 1$.

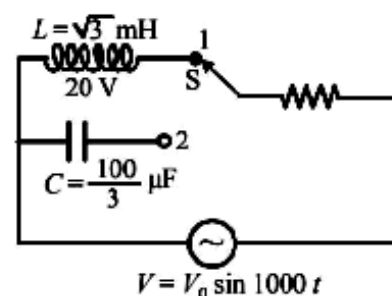


23. In the circuit shown, the value of L is 5 henry and the power factor of the circuit is 0.8. It is also given that the voltage drop across capacitor is $\frac{2}{5}$ times the voltage drop across the inductor. Find impedance (in ohm) of the circuit.



24. Two bodies of same mass tied with an inelastic string of length ℓ lie together. One of them is projected vertically upwards with velocity $\sqrt{6g\ell}$. The maximum height up to which the centre of mass of system of the two masses rises is $k\ell$. Find the value of k .

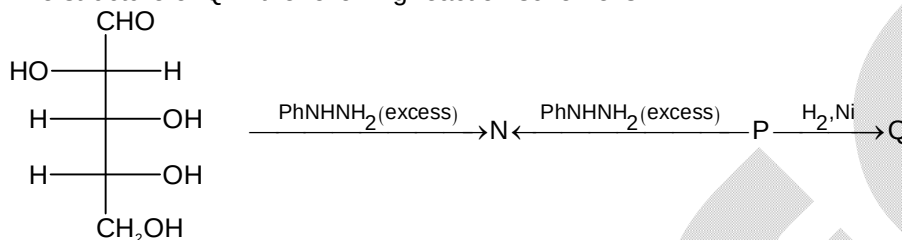
25. A metal rod AB of length $10x$ has its one end A in ice at 0°C and the other end B in water at 100°C . If a point P on the rod is maintained at 40°C , then it is found that equal amounts of water and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g . If the point P is at a distance of λx from the ice end A. The value of λ is (Neglect any heat loss to the surroundings.)
26. A dielectric disc has uniform charge density δ only in shaded portion. Now the electric field intensity at point P is equal to (If $\delta = 20\epsilon_0\text{ SI unit}$, $d = \frac{4R_1}{3}$, $R_2 = \frac{16R_1}{9}$)
27. A particle starts from rest at the origin and moves along the x -axis under the action of a force $F = (12 - 2t)$ Newton. Find the time after which it will be at the origin again.
28. An α -particle is accelerated by a potential difference of 10^4V . Find the change in its direction (in degree) of motion as angle measured in degree if it enters normally in a region of thickness 0.1 m having transverse magnetic induction of 0.1 T [given mass of α -particle = $6.4 \times 10^{-27}\text{ kg}$].
29. A glass rod of diameter $d_1 = 1.5\text{ mm}$ is inserted symmetrically into a glass capillary of inside diameter $d_2 = 2\text{ mm}$. Then the whole arrangement is vertically oriented and brought in contact with the surface of water. To what height (in mm) will the water rise in the capillary? (Density of water = 1 gm/cc , surface tension = 0.07 N/m , angle of contact = 0° , $g = 10\text{ m/s}^2$)
30. In the given AC circuit, when switch S is at position 1, the source emf leads current by $\frac{\pi}{6}$. Now, if the switch is at position 2, then source emf leads current by $\frac{\pi}{x}$. Find the value of x .



Chemistry**PART – B****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The structure of Q in the following reaction scheme is

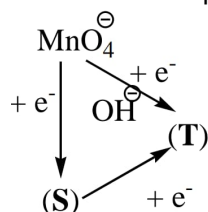


M(enantiopure)

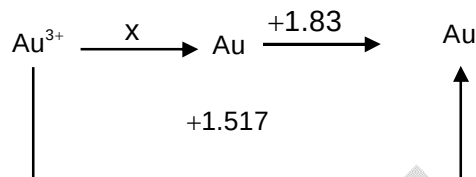
- (A) $\begin{array}{c} \text{CHO} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$
- (B) $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$
- (C) $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CH}_2\text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$
- (D) $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{CH}_2\text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{CH}_2\text{OH} \end{array}$

32. The magnetic moment (in units of BM) of copper in $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$ respectively is
 (A) 1.73 and 0
 (B) 1.73 and 1.73
 (C) 2.83 and 2.83
 (D) 0 and 2.83
33. The transition metal (M) complex that can have all isomers (geometric, linkage, and ionization) is
 (A) $[\text{M}(\text{NH}_3)_4\text{Br}_2]\text{SCN}$
 (B) $[\text{M}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$
 (C) $[\text{M}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_3$
 (D) $[\text{M}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{SCN})$
34. The products X and Y in the following reaction sequence, respectively, are
 $\text{BCl}_3 \xrightarrow[150^\circ\text{C}]{\text{NH}_4\text{Cl in C}_6\text{H}_5\text{Cl}} \text{X} \xrightarrow{\text{NaBH}_4} \text{Y}$
 (A) $\text{B}_3\text{N}_3\text{Cl}_6$ and $\text{B}_3\text{N}_3\text{H}_6$
 (B) $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3$ and $\text{B}_3\text{N}_3\text{H}_6$
 (C) $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3$ and $\text{B}_3\text{N}_3\text{H}_{12}$
 (D) $\text{B}_3\text{N}_3\text{H}_9\text{Cl}_3$ and $\text{B}_3\text{N}_3\text{H}_{12}$

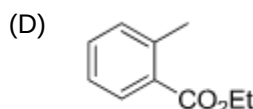
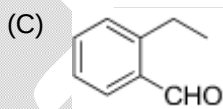
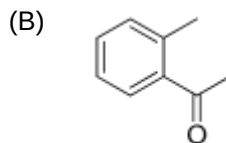
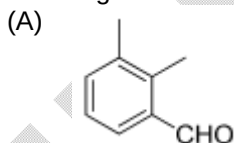
35. Following is the reaction flow chart for manganese oxidocomplexes under different alkaline pH conditions. Compounds (S) and (T) respectively are



- (A) S = $\text{MnO}(\text{OH})_2$; T = $\text{Mn}(\text{OH})_2$
 (B) S = MnO_2 ; T = $\text{MnO}(\text{OH})$
 (C) S = MnO_4^{2-} ; T = $\text{MnO}(\text{OH})$
 (D) T = MnO_4^{2-} ; S = MnO_2
36. Number of different types of dipeptides produced using a mixture of glycine and L-valine, and number of optically active dipeptides formed in this mixture will be
 (A) Four dipeptides, all optically active
 (B) Two dipeptides, all optically active
 (C) Four dipeptides, three optically active
 (D) Two dipeptides, none optically active
37. Latimer diagram are the compact representations of electrochemical equilibria in substances of multiple oxidation states. The value of the potential, x, in the Latimer diagram of gold (at pH = 1.0) is



- (A) 2.72 V
 (B) 3.18 V
 (C) -3.18
 (D) 1.36 V
38. An organic compound on reaction with 2,4-dinitrophenylhydrazine (2,4-DNP) gives a yellow precipitate. It also gives silver mirror on reaction with ammoniacal AgNO_3 . It gives an alcohol and sodium salt of a carboxylic acid on reaction with concentrated NaOH . It yields benzene-1,2-dicarboxylic acid on heating with alkaline KMnO_4 . The structure of the compound among the following is



39. The pH of an aqueous buffer prepared using CH_3COOH and $\text{CH}_3\text{COO}^- \text{Na}^+$ is 4.80.

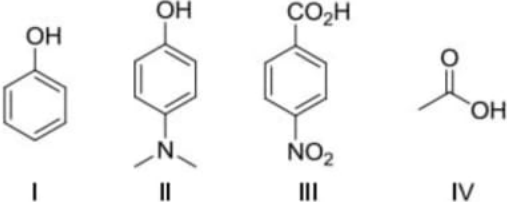
The quantity $\frac{[\text{CH}_3\text{COO}^-] - [\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COOH}]}$ is _____

(round off to three decimal places)

[Given: pK_a of CH_3COOH in water is 4.75]

Given $10^{0.05} = 1.1220$

- (A) 0.122
 (B) 0.125
 (C) 0.120
 (D) 0.123

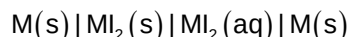
40. The molar heat capacity of a substance is represented in the temperature range 298 K to 400 K by the empirical reaction $C_{p,m} = 14 + bT \text{ J K}^{-1} \text{ mol}^{-1}$, where b is a constant. The molar enthalpy change when the substance is heated from 300 K to 350 K is 2 kJ mol^{-1} . The value of b nearest to _____ $\text{J K}^{-2} \text{ mol}$
- (A) 0.08 (B) 0.12
(C) 0.05 (D) 0.10
41. The observed magnetic moment of octahedral Mn^{3+} , Fe^{3+} and Co^{3+} complexes are 4.95, 6.06 and 0.00 BM, respectively. The correct option for the electronic configuration of Mn^{3+} , Fe^{3+} and Co^{3+} metal ion in these complexes, respectively, is
- (A) $t_{2g}^4 e_g^0, t_{2g}^3 e_g^2$ and $t_{2g}^4 e_g^2$ (B) $t_{2g}^3 e_g^1, t_{2g}^5 e_g^0$ and $t_{2g}^6 e_g^0$
(C) $t_{2g}^3 e_g^1, t_{2g}^3 e_g^2$ and $t_{2g}^6 e_g^0$ (D) $t_{2g}^2 e_g^1, t_{2g}^3 e_g^2$ and $t_{2g}^4 e_g^2$
42. The complementary strand for the following single strand of DNA is
- 5' ← A — T — G — C — T → 3'
- (A) 3' ← T — A — C — G — A → 5'
(B) 3' ← A — T — G — C — T → 5'
(C) 5' ← T — A — C — G — A → 3'
(D) 5' ← A — A — C — G — T → 3'
43. The correct order of pK_a for the following compounds is
- 

 I II III IV
- (A) II > I > III > IV (B) II > I > IV > III
(C) III > IV > I > II (D) IV > II > I > III
44. The rate constant values for the decay of radioisotopes X and Y, used in ratio-medicine are 0.05 h^{-1} and 0.025 h^{-1} respectively. In a hospital, at a time ' t_0 ' the activity of a sample of X was found to be twice that of Y. The activities of the two radioisotopes will be approximately equal when the time elapsed is:
- (A) twice the half-life of Y (B) twice and half-life of X
(C) equal to the half-life of X (D) equal to $1/2$ the half-life of Y
45. Sea water containing 1 M NaCl has to be desalinated at 300 K using a membrane permeable only to water. The minimum pressure (in bars) required on the sea-water side of the membrane is _____
- ($R = 8.3 \text{ J mol}^{-1} \text{ K}^{-1}$, $1 \text{ bar} = 10^5 \text{ N/m}^2$)
- (A) 24.94 (B) 49.88
(C) 2490 (D) 4980
46. The property of radiation that is not different at various regions of the electromagnetic spectrum is
- (A) energy (B) frequency
(C) velocity (D) wavelength

47. Among the following species, the one that has pentagonal shape is

(A) XeOF_4 (B) IF_5
(C) $[\text{SF}_5]^-$ (D) $[\text{XeF}_5]^-$

48. Consider the electrochemical cell



where 'M' is a metal. At 298 K, the standard reduction potential are

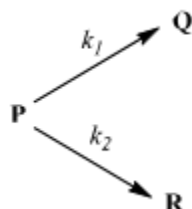
$E_{\text{M}^{2+}(\text{aq})/\text{M(s)}}^0 = -0.12 \text{ V}$, $E_{\text{Ml}_2(\text{s})/\text{M(s)}}^0 = -0.36 \text{ V}$ and the temperature coefficient is

$\left(\frac{\partial E_{\text{cell}}^0}{\partial T}\right)_p = 1.5 \times 10^{-4} \text{ V K}^{-1}$. At this temperature the standard enthalpy change for the overall cell

reaction $\Delta_r H^0$, is _____ kJ mol^{-1}

(A) -37.60 (B) 37.60
(C) 3.76 (D) -3.76

49. Consider the following two parallel irreversible first order reactions at temperature T



where k_1 and k_2 are the rate constants and their values are 5×10^{-2} and $15 \times 10^{-2} \text{ min}^{-1}$, respectively, at temperature T. If the initial concentration of the reactant 'P' is 4 mol L^{-1} , then the concentration of product 'R' after 10 min of reaction is _____ mol L^{-1} .

Given $e^{-2} = 0.135$

(A) 1.59 (B) 2.59
(C) 3.59 (D) 4.59

50. Gas phase bond length and dipole moment of a compound (MX) is $3 \text{ \AA}$ and 10.8 D , respectively. The ionic character in gas phase MX is _____ %.

($1\text{D} = 3.336 \times 10^{-30} \text{ C m}$)

(A) 64.50 (B) 74.50
(C) 84.50 (D) 90.50

SECTION – B

(Numerical Answer Type)

This section contains 10 Numerical based questions. The answer to each question is rounded off to the nearest integer value.

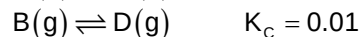
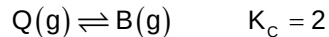
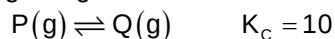
51. The total number of α and β particles emitted in the following radioactive decay is x. Find the

value of $\frac{x \times 5}{11}$.



52. For the elementary reaction $\text{C} \xleftarrow{k_2} \text{A} \xrightarrow{k_1} \text{B}$, $k_1 = 2k_2$. At time $t = 0$, $[\text{A}] = A_0$ and $[\text{B}] = [\text{C}] = 0$. At a later time t, the value of $[\text{B}]/[\text{C}]$ is _____ (round off to the nearest integer)

53. For the given general reactions



Calculate the value of K_c for the reaction $D(g) \rightleftharpoons P(g)$

54. The number of unpaired electron in $[Co(H_2O)_6]^{2+}$ is _____
55. The volume of water (in mL) required to be added to a 100 mL solution (aq. 0.1 M) of a weak acid (HA) at 25°C to double its degree of dissociation is x, find the value of $x/100$
[Given: K_a of HA at 25 °C = 1.8×10^{-5}]
56. The ratio of the $2p \rightarrow 1s$ transition energy in He^+ to that in the H atom is closest to
57. Total number of paramagnetic species NO, O_2 , B_2 and C_2 in their ground state is
58. Half-life ($t_{1/2}$) of a chemical reaction varies with the initial concentration of reactant (A_0) as given below

A_0 (mol L ⁻¹)	5×10^{-2}	4×10^{-2}	3×10^{-2}
$t_{1/2}$	360	450	600

The order of the reaction is _____

59. For the reaction
 $H_2PO_2^-(aq) + OH^-(aq) \rightarrow HPO_3^{2-}(aq) + H_2(g)$
 the rate expression is $k[H_2PO_2^-][OH^-]^2$. If the concentration of $H_2PO_2^-$ is doubled, the new rate is _____ $\times$ old rate
60. On heating a sample of 25 mg hydrated compound (molecular weight = 250 g/mol) in thermogravimetric analysis, 16 mg of dehydrated compound remains. The number of water molecules lost per molecule of hydrated compound is _____.
 (Molecular weight of water = 18 g/mol)

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The value of $\sin^{-1} \left\{ \cot \left(\sin^{-1} \sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1} \frac{\sqrt{12}}{4} + \sec^{-1} \sqrt{2} \right) \right\}$ is equal to
 (A) 1 (B) 0
 (C) -1 (D) 2
62. If $\frac{\tan 3\theta}{\tan \theta} = 4$, then $\frac{\sin 3\theta}{\sin \theta}$ equals
 (A) $\frac{3}{5}$ (B) $\frac{4}{5}$
 (C) $\frac{3}{4}$ (D) none of these
63. Let $A = \{6, 10, 14, \dots, 1002\}$ and B is set of divisors of the integer 360, then $S(A \cap B)$ is [S(A) denotes sum of elements in set A]
 (A) 148 (B) 154
 (C) 156 (D) 162
64. Number of integral values of λ for which $f(x) = \sqrt{\ln(2\lambda \cos x + 5)}$ is defined for all $x \in \mathbb{R}$ is –
 (A) 5 (B) 4
 (C) 6 (D) 11
65. The solution of the differential equation $\{y(1+x^{-1}) + \sin y\}dx + (x + \ln x + x \cos y)dy = 0$ is –
 (A) $x + y \ln x + y \sin x = c$ (B) $xy + y \ln x + x \sin y = c$
 (C) $y + x \ln x + x \sin y = c$ (D) $xy + x \ln x + y \sin y = c$
 (where c is arbitrary constant)
66. If in triangle ABC, $A \equiv (1, 10)$, circumcentre $\equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocenter $\equiv \left(\frac{11}{3}, \frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is
 (A) $\left(1, -\frac{11}{3}\right)$ (B) (1, 5)
 (C) (1, -3) (D) (1, 6)
67. Consider a circle with centre lying on the focus of the parabola $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle and the parabola is
 (A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -\frac{p}{2}\right)$
 (C) $\left(-\frac{p}{2}, p\right)$ (D) $\left(-\frac{p}{2}, -p\right)$

68. If the range of values of a such that a circle passes through the points of intersection of $y = x^2$ and $x^2 - y^2 = a^2$ is (α, β) then $|\alpha - \beta|$ equals
 (A) 0 (B) 1
 (C) 2 (D) 3
69. Let $\vec{\alpha} = a\hat{i} + b\hat{j} + c\hat{k}$ and $\vec{\beta} = b\hat{i} + c\hat{j} + a\hat{k}$, where $a, b, c \in \mathbb{R}$. If ' θ ' be the angle between $\vec{\alpha}$ and $\vec{\beta}$ then,
 (A) $\theta \in (0, \pi/2)$ (B) $\theta \in [0, 2\pi/3]$
 (C) $\theta \in (2\pi/3, \pi]$ (D) none of these
70. If $a, b, c > 0$ and $x, y, z \in \mathbb{R}$, then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$ is equal to
 (A) $a^x + b^y + c^z$ (B) $a^{-x} b^{-y} c^{-z}$
 (C) $a^{2x} b^{2y} c^{2z}$ (D) 0
71. A and B are two events such that $P(A) = 0.2$ and $P(A \cup B) = 0.7$. If A and B are independent events then $P(B')$ equals
 (A) $2/7$ (B) $7/9$
 (C) $3/8$ (D) none of these
72. The value of $\sum_{r=1}^{1024} [\log_2 r]$ is equal to, ($[.]$ denotes the greatest integer function)
 (A) 8192 (B) 8204
 (C) 8194 (D) none of these
73. Number of natural numbers $< 2 \cdot 10^4$ which can be formed with the digits 1, 2, 3 only is equal to
 (A) $\frac{3^5 + 2 \cdot 3^4 - 3}{2}$ (B) $\frac{3^6 - 2 \cdot 3^4 + 3}{2}$
 (C) $\frac{3^7 - 1}{2}$ (D) none of these
74. The value of $i \log(x - i) + i^2 \pi + i^3 \log(x + i) + i^4 (2 \tan^{-1} x)$, $x > 0$ (where $i = \sqrt{-1}$) is
 (A) 0 (B) 1
 (C) 2 (D) 3
75. If $3x^2 - 2(a - d)x + (a^2 + 2(b^2 + c^2) + d^2) = 2(ab + bc + cd)$, then
 (A) a, b, c, d are in G.P. (B) a, b, c, d are in H.P.
 (C) a, b, c, d are in A.P. (D) None of these
76. If $a \sin x + b \cos(C + x) + b \cos(C - x) = \alpha$, then the minimum value of $|\cos C|$ is
 (A) $\sqrt{\frac{\alpha^2 - a^2}{b^2}}$ (B) $\sqrt{\frac{\alpha^2 - a^2}{4b^2}}$
 (C) $\sqrt{\frac{\alpha^2 - a^2}{16b^2}}$ (D) none of these

77. The solution of the differential equation $ydx + xy^2 dx = xdy$, is
 (A) $\frac{x}{y} + x^2 = \lambda$ (B) $\frac{x}{y} + \frac{x^2}{2} = \lambda$
 (C) $\frac{x}{2y^2} + \frac{x^2}{4} = \lambda$ (D) None of these
78. The value of $\int_0^{2\pi} [|\sin x| + |\cos x|] dx$ is equal to where $[.]$ represents GIF
 (A) $\frac{\pi}{2}$ (B) π
 (C) $\frac{3\pi}{2}$ (D) 2π
79. If $3(a + 2c) = 4(b + 3d)$ then the equation $ax^3 + bx^2 + cx + d = 0$ will have
 (A) no real solution (B) atleast one real root in $(-1, 0)$
 (C) atleast one real root in $(0, 1)$ (D) none of these
80. The mean and variance of seven observations are 8 and 16, respectively. If 5 of the observations are 2, 4, 10, 12, 14, then the product of the remaining two observations is :
 (A) 40 (B) 45
 (C) 49 (D) 48

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. If $f(x) = \int_0^x \frac{dt}{(f(t))^2}$ and $\int_0^2 \frac{dt}{(f(t))^2} = (6)^{\frac{1}{3}}$ then $f(9)$ is equal to
82. Consider an ellipse, whose centre is at the origin and its major axis is along the x-axis. If its eccentricity is $\frac{3}{5}$ and the distance between it foci is 6, then the area (in sq. units) of the quadrilateral inscribed in the ellipse, with the vertices as the vertices of the ellipse, is :
83. If M, N are 3×2 , 2×3 matrices such that $MN = \begin{bmatrix} 8 & 2 & -2 \\ 2 & 5 & 4 \\ -2 & 4 & 5 \end{bmatrix}$, then $\left[\frac{1}{10} \cdot \det(NM) \right]$ is equal to ____
 (NM is invertible and $[.]$ denotes the greatest integer function)
84. Number of positive integers n less than 15, for which $n! + (n+1)! + (n+2)!$ is an integral multiple of 49, is
85. The sum of coefficients of even powers of x in the expansion of $\left(x + \frac{1}{x} \right)^{11}$ is

86. The number of points of discontinuity of $f(x)$ in $[0, 2]$ where
$$f(x) = \begin{cases} [\cos \pi x], & x \leq 1 \\ |2x - 3| [x - 2], & x > 1 \end{cases}$$
 ($[.]$ denotes the greatest integer function), are
87. The curve $x + y - \ln(x + y) = 2x + 5$ has a vertical tangent at the point (α, β) . Then $\alpha + \beta$ is equal to
88. Let $A = \begin{bmatrix} x^2 & 6 & 8 \\ 3 & y^2 & 9 \\ 4 & 5 & z^2 \end{bmatrix}$, $B = \begin{bmatrix} 2x & 3 & 5 \\ 2 & 2y & 6 \\ 1 & 4 & 2z - 3 \end{bmatrix}$. If $\text{trace } A = \text{trace } B$ then $x + y + z$ is equal to
(trace equal to sum of principal diagonal elements in a matrix)
89. The area enclosed by the curve $|y| = \sin 2x$, when $x \in [0, 2\pi]$ is
90. If the length of shortest distance the two lines $\frac{1}{2}(x - 1) = \frac{1}{4}(y - 3) = z + 2$ and $3x - y - 2z + 4 = 0$
 $= 2x + y + z + 1$ is $s\sqrt{14}$. Then the value of s is _____